

BADDEL LAUNCHER

Technical Reference — v1.0
Last updated: March 2026

PostHog	GA4 Realtime	Consent-first
Behavioural Events	Active Users Only	Privacy by Design

1. Overview

Baddel Launcher uses **two separate analytics systems**, each with a distinct purpose. They share the same anonymous installation identifier so that data across both tools maps to the same device — without ever collecting personal information.

Tool	Purpose	Data Sent	Consent Required?
PostHog	Behavioural events — understand how users interact with features	Bucketed counts, platform names, feature flags	✓ Yes (opted-in only)
Google Analytics 4	Realtime active-users dashboard only	session_id, country code, app version	✗ No — essential service ping

2. Shared Installation ID

Both PostHog and GA4 use the same **installation_id** — a random UUID generated on first launch and persisted locally. It is never linked to a name, email, or any account. It exists solely to deduplicate events from the same device.

File location	%APPDATA%\BaddelLauncher\installation_id.txt
Format	UUID v4 e.g. a3f8c2d1-...
Lifecycle	Created once; never rotated; deleted with app data
Shared with	PostHog (distinct_id) • GA4 (client_id)

3. PostHog — Behavioural Events

PostHog receives detailed behavioural events that help the team understand feature adoption, session lengths, and library health. **These events are only sent after the user explicitly grants consent.** The one exception is the essential ping described in section 3.3.

3.1 Endpoint

Host	us.i.posthog.com
Path	/batch/
Method	POST • Content-Type: application/json
Batching	Up to 50 events per request; flushed every 60 s or on queue fill
Offline queue	%APPDATA%\BaddelLauncher\analytics_queue.json — events survive restarts

3.2 Event Reference

Event Name	Trigger	Properties Sent	Consent?
launcher_ping	App starts	status: "alive"	No — essential
app_launched	App fully loaded (after games + collections loaded)	platforms_connected, library_size_bucket, collections_count	Yes
game_launched	User clicks Play on any game	platform, source	Yes
game_session_ended	Game process exits	platform, duration_bucket	Yes
game_added_manual	User adds a game manually via file picker	—	Yes
game_removed	User hides/removes a game	platform	Yes
game_restored	User restores hidden game(s)	count	Yes
game_deleted_forever	User permanently deletes a game	—	Yes
game_spin_clicked	Random game picker used	custom_pool (bool)	Yes
game_image_changed	Cover or hero image updated	image_type, is_reset	Yes
library_scanned	Full library scan completes	games_count_bucket, platforms_detected	Yes
account_added	New account saved for any platform	platform	Yes
account_switched	Active account switched	platform	Yes
add_account_clicked	Add Account button pressed	platform	Yes
account_deleted	Account deleted	platform	Yes

Event Name	Trigger	Properties Sent	Consent?
collection_created	New collection created	is_system (bool)	Yes
game_added_to_collection	Game added to any collection	is_favorite (bool)	Yes
game_removed_from_collection	Game removed from collection	is_favorite (bool)	Yes
hud_sensor_toggled	HUD overlay sensor enabled / disabled	enabled (bool)	Yes
feedback_sent	In-app feedback submitted	—	Yes
opted_in	User grants analytics consent	—	Yes

3.3 Essential Ping (No Consent Required)

The `launcher_ping` event is sent every time the app starts regardless of consent status. It carries only *status*: "alive" and the app version. It does not include OS, game data, or any user action. Its purpose is to give the team a rough install count. The OS field is explicitly set to "hidden" for this event.

3.4 Common Properties on Every Event

distinct_id	Installation UUID (anonymous)
time	Unix timestamp (seconds)
\$app_version	Semver string e.g. "1.2.0"
\$os	"win10" "win11" "win_other" "unknown" — hidden on essential events
\$lib	"Baddel-Launcher-Electron"

3.5 Bucketing — No Raw Numbers

Numeric values (game counts, session durations) are never sent as raw numbers. They are mapped to fixed buckets before sending, making it impossible to infer exact playtime or library size for any user.

Type	Raw Value	Bucket Sent
Session duration	7 minutes	"5-15m"
Session duration	95 minutes	"1-2h"
Count (games, etc.)	34 games	"25-50"
Count (games, etc.)	3 games	"1-5"

4. Google Analytics 4 — Realtime Active Users

GA4 is used for a single purpose: showing how many instances of Baddel Launcher are currently open in the GA4 Realtime dashboard. No behavioural events, no game data, and no account information are ever sent to GA4.

4.1 Endpoint

Host	www.google-analytics.com
Path	/mp/collect?measurement_id=G-7C9JNWLNVZ&api_secret=...
Method	POST • Content-Type: application/json
Protocol	GA4 Measurement Protocol (server-side)

4.2 Events Sent

Event Name	Frequency	Purpose
app_open	Once on launch	Marks the user as active in the realtime view
user_engagement	Every 2 minutes	Keeps the user visible in GA4's 5-min sliding realtime window

4.3 Payload Structure

The following fields are included in every GA4 request. No other data is attached.

client_id	Installation UUID (same as PostHog distinct_id)
session_id	Unix timestamp at app launch — resets every time the app opens
engagement_time_msec	Always 100 ms — GA4 requirement, not real measurement
user_country	ISO country code from api.country.is (e.g. "EG", "US"). Falls back to "Unknown"
app_version	Semver string e.g. "1.2.0"

4.4 Why No Consent Gate?

The GA4 ping is functionally equivalent to a connection test — it tells the server that the app is running, not what the user is doing. It carries no game names, no account data, and no behavioural signal. It is treated as an essential telemetry ping (similar to auto-update checks) and therefore does not require explicit opt-in.

5. Consent System

5.1 Storage

File	%APPDATA%\BaddelLauncher\analytics_consent.txt
Values	"true" "false"
Default	"false" — analytics is OFF until the user explicitly opts in

5.2 What Changes With Consent

Behaviour	Consent = false (default)	Consent = true
launcher_ping	Sent	Sent
All other events	Blocked	Sent
\$os field	"hidden"	win10/11/...
GA4 heartbeat	Sent	Sent
Offline queue	Not written	Written

5.3 IPC API (Renderer → Main)

IPC Channel	Action
analytics-grant-consent	Sets consent = true, saves file, fires opted_in event
analytics-revoke-consent	Sets consent = false, saves file, stops future events
analytics-is-enabled	Returns current consent boolean (sync)
analytics-log-hud-sensor	Logs hud_sensor_toggled (guarded by consent)
analytics-log-game-spin	Logs game_spin_clicked (guarded by consent)
analytics-log-image-changed	Logs game_image_changed (guarded by consent)
analytics-log-feedback	Logs feedback_sent (guarded by consent)

6. Data Flow Summary

S te p	Actor	Action
1	App starts	analytics.init() runs — loads installation_id, consent flag, offline queue
2	PostHog	launcher_ping fired immediately (essential, no consent check)
3	GA4	Country fetched from api.country.is → app_open fired → 2-min heartbeat starts

Step	Actor	Action
4	User action	Event written to in-memory queue, then flushed to disk (analytics_queue.json)
5	Flush timer	Every 60 s: up to 50 queued events are batch-sent to PostHog /batch/
6	GA4 timer	Every 2 min: user_engagement sent to keep user in GA4 realtime window
7	App closes	Unsaved playtime written; queue persists on disk for next launch

7. Local Files Created by Analytics

File	Contents	Deletable?
installation_id.txt	Random UUID, one line	Yes — new ID generated on next launch
analytics_consent.txt	"true" or "false"	Yes — defaults to false
analytics_queue.json	Array of pending PostHog events	Yes — lost events discarded

8. Privacy Principles

No PII collected	No names, emails, Steam IDs, account usernames, or file paths are ever sent to any analytics endpoint.
No raw numbers	Game counts and session durations are bucketed before transmission. Exact values stay on-device.
Consent-first for events	All behavioural events are blocked until the user actively opts in. The default state is off.
Minimal essential data	The only data sent without consent is launcher_ping (status: alive) and the GA4 realtime ping, both carrying no user-action data.
Single anonymous ID	One UUID ties both analytics tools to the same device — no cross-referencing with external identity systems.
Offline-resilient queue	Events are persisted locally and retried on next launch. No data is silently lost, and no duplicate sends occur.

OS detection once	The OS version is detected a single time at startup using a PowerShell command. It is cached and never re-queried.
Fire-and-forget GA4	GA4 requests do not await a response and errors are silently swallowed — GA4 outages cannot affect app stability.

This document reflects the analytics implementation in analytics.js and main.js as of the March 2026 production build. Any future changes to event schemas or consent logic should be reflected here.